



# Tiago Barros

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Tiago Barros

## Skills

### Networking

IPv4, IPv6, Troubleshooting, VLANs, Routing

### Linux

Ubuntu, Debian, Bash, Systemd

### Orchestration

Kubernetes, Helm, Argo CD, Docker Compose

### Virtualization

Proxmox, OpenStack, Snapshots, Backups,  
Software-Defined Networking

### Monitoring & Observability

Prometheus, Grafana, Loki, Telegraf, InfluxDB

### Programming

Go, Python, Bash, C/C++, Java

### Git & GitHub

Code Review, Pull Requests

### Cloud-Native

GitOps, Docker, LXC Containers

## Languages

### English

Highly Proficient

### Portuguese

Native

## Experience

### University of Aveiro

May, 2024 - Present

Research & Infrastructure Engineer

Aveiro

- Build and maintain **infrastructure** for research and production, focusing on **networking, virtualization, and automation**.
- Deploy and manage **Proxmox** and **OpenStack** environments, IP Networks, and observability stacks (**Telegraf, InfluxDB, Grafana, Prometheus**).
- Support advanced **5G network** projects, integrating slice orchestration, REST APIs, and energy-aware networking.

### Institute of Telecommunications

May, 2023 - May, 2024

Research Fellow

Aveiro

- Researched and deployed **multi-edge, multi-slice private 5G** networks with the Aether platform.
- Developed a **GitOps**-driven CLI tool for slice deployment and configuration.
- Authored a **peer-reviewed paper** on flexible 5G slice provisioning, integrating open-source RAN solutions.

## Education

### University of Aveiro

Sep, 2018 - Dec, 2023

Computer Engineering and Telematics

Master's

Thesis Title: "*Dynamizing Private 5G Edge Deployments*"

## Projects

### Homelab

Oct, 2024 - Present

Platform/SRE & Networking Sandbox

Built and run a Proxmox-based homelab: OpenWRT-routed dual-stack (**IPv4/IPv6**) with VLANs, WireGuard **VPN** for remote access, **Nginx** as a reverse proxy with **TLS (Certbot + Cloudflare DNS)** and fail2ban; observability via **Prometheus + Loki + Grafana**.

### Private Multi-Edge 5G Network

Oct, 2022 - May, 2024

(Aether/SD-Core + GitOps + CLI)

**Researched** and **validated** a standalone private 5G network across multiple edge sites employing different network slices using Aether & SD-Core; built a **CLI** to streamline network operations and developed an **API** for higher-level operations; Deployments supported via GitOps (**Argo CD + GitHub**).

### Programming Projects

2020 - Present

Go, Python, C/C++, Java

- ETF Knapsack (*Python*) – Optimizer for ETF portfolio allocations under budget and weight constraints.
- Advent of Code (*Go*) – Solutions to annual programming puzzles, focusing on idiomatic Go.
- Lane Merge Simulator (*Python + Flask*) – ITS-G5-based vehicle merge orchestration simulation with web UI.
- Sokoban Solver (*Python*) – AI pathfinding (A\*) to solve arbitrary Sokoban puzzle layouts.